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End-of-life antibiotic stewardship: perspectives from the ESCMID Study Groups for Antimicrobial Stewardship and Infections in the Elderly

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Antibiotics are among the most frequently prescribed medications in end-of-life care, a period when patients live with progressive, life-limiting conditions and death is anticipated within weeks to months, shifting care priorities from curative to comfort-focused interventions. Evidence shows that up to 80% of patients in hospice or palliative care settings receive antibiotics in their final weeks, often with limited clinical benefit (1). While antibiotics can occasionally relieve symptoms, such as dysuria from urinary tract infections or pain and malodor from skin infections, they may also cause harm, including adverse drug reactions, *Clostridioides difficile* infections, intravenous line complications, or burdensome medicalization. Unlike other potentially non-beneficial end-of-life medications, antibiotics also carry a broader societal impact by driving antimicrobial resistance and facilitating the spread of multidrug-resistant organisms.

Despite their frequent use, significant uncertainties and ethical dilemmas persist regarding the role of antibiotics at the end-of-life, particularly in balancing potential symptomatic benefit, proportionality of care, patient-centered goals, and antimicrobial stewardship. This commentary aims to synthesize available evidence and expert perspectives from the ESCMID Study Groups for Antimicrobial Stewardship (ESGAP) and for Infections in the Elderly (ESGIE) to support clinicians in navigating antibiotic decision-making in end-of-life care.

Recognising the end-of-life and aligning goals of care

A recurring challenge in end-of-life antibiotic stewardship is timely recognition of patients approaching the end-of-life and accurate prognostication. Antibiotic decisions differ substantially depending on whether a patient is expected to live months, weeks, or only days to hours. Failure to recognize impending death may lead to reflexive escalation of antimicrobial therapy that is misaligned with patient values and goals of care.

Aligning antibiotic decisions with patient priorities requires early, values-based conversations. Multidisciplinary input is particularly important, as decisions around antibiotic initiation or discontinuation are rarely purely medical. They are influenced by family dynamics, prognostic uncertainty and emotional drivers such as fear, guilt or desire to “do everything possible”. Palliative care teams – multidisciplinary clinical teams, social workers and spiritual care professionals – are uniquely positioned to facilitate these conversations, address existential concerns and support shared decision-making by helping patients and families clarify values, preferences, and goals of care. Where formal palliative care is unavailable, collaboration among treating clinicians can still help clarify goals and reduce non beneficial treatment (1).

At the end-of-life, antibiotics should be considered primarily for their potential to improve symptoms rather than to prolong life. Evidence suggests that symptomatic benefit is inconsistent and highly context dependent. While some patients may experience relief from urinary symptoms or localized skin infections, antibiotics rarely improve symptoms related to respiratory secretions or terminal infections (1). Potential harms must be weighed carefully, particularly in frail patients. Adverse drug reactions, drug–drug interactions, *Clostridioides difficile* infection, and complications related to intravenous access may outweigh any marginal benefit. Antibiotic use may also contribute to unwanted medicalization, hospital transfers, or invasive procedures that conflict with comfort-oriented care.

When uncertainty persists, time-limited trials offer a pragmatic compromise. In this approach, antibiotics are prescribed with predefined goals (e.g., symptom relief, fever resolution, improved alertness) and a clear timeframe for reassessment. If these goals are not met, antibiotics are discontinued and care refocuses on comfort (2). Such strategies allow clinicians to address family concerns while preventing prolonged, non-beneficial antibiotic use.

Role of multidisciplinary teams and palliative care

Effective end-of-life antibiotic stewardship requires multidisciplinary collaboration. Decisions about antibiotic initiation or discontinuation, as well as administration options involve not only infection

management but also symptom control, prognostication, communication with families, and ethical considerations—dimensions that no single discipline can address alone.

Palliative care teams play a central role in this process. Retrospective studies from Korea show that patients with advanced cancer who received palliative care consultation were less likely to undergo escalation or combination therapy, and more likely to have antibiotics de-escalated or discontinued (3). Notably, these effects occurred despite the absence of direct antimicrobial stewardship recommendations, suggesting that reframing care around comfort, quality of life, and advance planning can indirectly shape prescribing behaviour (3).

However, most studies focus on antibiotic use in the final days of life. Whether earlier palliative care integration -weeks or months before death- could further optimize antimicrobial decision-making and stewardship remains largely unexplored. Earlier multidisciplinary engagement may offer greater opportunities to align treatment with patient values, address emotional drivers of escalation, and prevent non-beneficial antibiotic use (1).

Infectious diseases consultations at the end-of-life

Infectious disease (ID) specialists are increasingly consulted for seriously ill patients, including those nearing the end-of-life. A recent US-based study involving over 60,000 inpatient ID consults showed that patients receiving ID input between 2014 and 2023 were increasingly comorbid and had high short-term mortality, with 7.5% dying during admission, and six-month mortality exceeding 19% (3). During the same period, ID consultations per 100 admissions nearly doubled (from 5.0 to 9.9) suggesting growing involvement of ID specialists in high-risk and terminal populations.

Historically, antibiotics have been called “the drugs of fear”, reflecting overuse driven by diagnostic uncertainty and clinician anxiety (5). In this context, ID consultations may themselves function as a “consult of fear”, often requested close to the end-of-life to reassure medical teams and families that “everything possible” is being done. However, ID specialists are also uniquely positioned to support stewardship by contextualizing microbiological findings, clarifying realistic benefits, and reinforcing proportional, goal-concordant care. Their role at the end-of-life should therefore extend beyond pathogen-focused decision-making to include ethical reflection and patient-centered stewardship.

Routes of administration and practical considerations

When antibiotics are considered appropriate in end-of-life care, route of administration presents both practical and ethical challenges (1). Oral therapy is preferred when swallowing is intact, as it is less invasive and easier to manage in home or hospice settings. Intravenous (IV) therapy, while common in hospitals, may be burdensome and is associated with line-related complications such as infection or thrombosis, that are often inconsistent with comfort-focused goals.

The subcutaneous (SC) route offers a safe and patient-centered alternative: less invasive, suitable for home or hospice, and pharmacokinetically comparable to IV for time-dependent beta-lactams regarding area under the curve and trough concentrations (6, 7). Ceftriaxone is the most-studied agent, but other antibiotics including temocillin, cefazolin, ertapenem, meropenem, piperacillin/tazobactam and amoxicillin/clavulanic acid demonstrated promise (6). Aminoglycosides, however, are not recommended due to risk of local toxicity.

Despite these advantages, SC antibiotics remain an off-label practice with heterogenous uptake across Europe: common in France, but rare or absent in the UK or Germany. Local safety concerns, such as pain or skin reactions, appear underreported or overlooked. Simple measures -using non-rigid catheters and infusion over at least 30 minutes – can mitigate many issues. Given its potential to support patient comfort and avoid burdensome interventions, SC antibiotic administration deserves greater recognition in research and stewardship discussions (6).

International perspectives and institutional guidance

Practice variation and uncertainty are further illustrated by a nationwide Italian physician survey involving 885 respondents (8). The survey revealed inconsistent management of end-of-life infections, particularly regarding respiratory secretions, antibiotic duration, and stopping rules. Care setting and route of administration strongly influenced decisions, while more than half of respondents—except those working in palliative care—reported managing cancer and non-cancer patients differently. Importantly, most institutions lacked specific guidance on antibiotic use at the end-of-life.

These findings mirror reports from other countries (9,10) and underscore the need for institutional policies and national guidance that support clinicians facing complex, emotionally charged decisions. Clear frameworks can reduce variability, enhance confidence, and promote consistent, patient-centered stewardship practices.

Conclusion and perspectives from ESGAP and ESGIE

Antibiotic use at the end-of-life sits at the intersection of stewardship, ethics, and palliative care. Decisions are rarely straightforward and often reflect emotional, cultural, and institutional pressures. Drawing on the above perspectives, experts from ESGAP and ESGIE propose a practical framework to guide clinicians (Figure 1). Core principles include early recognition of the end-of-life, proactive palliative care involvement, use of time-limited trials, engagement of ID specialists as stewardship partners, optimization of administration routes to support comfort, and development of clear institutional and national guidance.

Antibiotic stewardship at the end-of-life requires more than microbiological expertise; it demands sensitive communication, multidisciplinary collaboration, and prioritization of patient dignity. By bringing together diverse perspectives, ESGAP and ESGIE aim to stimulate further discussion and research on how best to use — and when to withhold — antibiotics in this most delicate phase of care.

Figure 1. Practical framework for end-of-life antibiotic use

KEY PRINCIPLES		PRACTICAL CONSIDERATIONS
1	Recognize end-of-life and impending death	Train clinicians to identify patients (cancer and non-cancer) nearing end-of-life and diagnose impending death.
2	Early palliative care involvement	Early and proactive collaboration helps align care with patient values, improves communication and reduces futile antibiotic use.
3	Consider time-limited trials	Predefine criteria for benefit (symptom relief, fever resolution); discontinue if unmet.
4	Empower ID specialists and stewardship teams	Ensure microbial indications are balanced with comfort and proportionality.
5	Optimize administration route	Avoid intravenous perfusion if not required: oral route should be prioritized if possible or subcutaneous route as alternative
6	Strengthen institutional and national guidance	Clear, context-specific policies on antibiotic use in end-of-life care are urgently needed to reduce variability and support clinicians in difficult decision-making.
7	Address knowledge gaps through further research	Better tools are needed to support prognostication, evaluate symptomatic benefit, and assess the wider impact of antibiotic use in palliative care.

ID: Infectious diseases; IV: Intravenous; SC: Subcutaneous.

Authors contribution

LC, NPS, RM, and DAO (ESGAP Executive Committee) together with NB (ESGIE Executive Committee) coordinated the development of the commentary and oversaw contributions from AS, CRB, DK, SHY, MF, MM, and NB, who provided expert input. LC synthesised all contributions and drafted

177 the initial version of the commentary. All authors critically reviewed, revised, and approved the
178 manuscript, and LC finalized the text for submission.

179 **Transparency declaration**

180 **Conflict of interest**

181 LC, DOA, NP, and RM are members of the ESCMID Study Group for Antimicrobial Stewardship
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